

Simulation Credence and Its Consequences

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January 2026

Abstract

I propose a two-level representation of candidate worlds as pairs (L, h) , where L fixes nomic constraints and h fixes a concrete realization. The framework distinguishes two routes to high simulation credence: a prevalence route, where probability of simulation hypotheses can be high conditional on dominance-type realizations, and a revelation route, where host-level intervention produces publicly checkable evidence E that makes simulation the uniquely best explanation from within w . I then argue that high simulation credence can have epistemic consequences in the two cases, affecting what we can know and how we should conduct inquiry.

1 Introduction

Two questions are often run together in simulation debates: how likely it is that we are simulated, and whether simulation could be established from within our world. This paper treats both questions within a single two-level framework, where a candidate world is a pair (L, h) of a law-profile L and a concrete realization h .

Using this setup, I contrast two routes to high simulation credence. The first is a prevalence route observer-counting conditional on ours-like laws and on realizations with many simulated observers, with aim to formalize Bostrom's and Chalmer's dominance argument and show its underdetermination without further assumptions about realization (necessary assumptions) probability. The second is a revelation route, where publicly checkable evidence produced by host-level intervention makes simulation the uniquely best explanation available from within w . The paper then briefly sketches how high simulation credence would affect inquiry by making apparent laws level-relative and potentially host-contingent, with different implications in the two cases.

2 Framework and notation

2.1 Possible worlds and their two-level structure

Let S be a set of candidate worlds (possible universes).

- 1st level: Each world $s \in S$ comes with a *law-profile* L_s : an idealized specification of the fundamental nomic constraints of s (“laws of nature”). Using Chalmers term “sim blockers”, L could for example contain impossibility of simulation, substrate (in)dependence, information/matter limitations and constraints, etc.
- 2nd level: h will then be a particular realization of some world. Given some law-profile L , an h fixes a full way the world could be under the laws of L : global structure, topology, initial conditions or developments including biological, cultural, and political trajectories, etc. In sense of Chalmers sim blockers, things like “nonsims will die out before creating sims” or “ethical reasons” would fall under h . H_L is then the set of all possible realizations of a given law-profile L .

From this two-level perspective, we then think of possible worlds as pairs:

$$s = (L, h),$$

where L is a *law-profile* (a set of nomic constraints) and h is its *realization* (a universe satisfying those constraints).

In short: for some possible world s , L fixes its nomic constraints, while h fixes a full concrete unfolding under them, an actual realization of the world.

2.2 Simulation and hosting

A *simulation* is an implemented, rule-governed process that generates an evolving structure of states intended to represent some world. Such simulation represents a world-like environment, represented by a pair (L, h) .

A *host world* is the world in which the simulation is realized. Host worlds may themselves be simulated, and can be represented by the same two-level notion (L, h)

A *simulated observer* is an entity whose life and cognitive functions are realized inside the simulation process.

We write $Host(s, t)$ for some possible worlds s and t iff world s contains an implemented simulation of the world t .

Fix our actual world as $w \in S$, where $w = (L_w, h_w)$. (The simulation hypothesis then can be reformulated as $\exists s \in S, Host(s, w)$).

3 Two scenarios of high simulation credence

3.1 Ours-like worlds and prevalence

To connect with Bostrom’s arguments, fix a minimal “ours-like” law core L_0 capturing the parts of our best physics we treat as non-negotiable for the discussion (the details about what particular laws are considered are left schematic here), i. e. $L_0 \subset L_w$. We call a world s *ours-like* iff its laws contain that core (and possibly some other new non-contradictory laws):

$$S_w := \{ s \in S \mid L_0 \subseteq L_s \}.$$

In other words, an ours-like world is one whose fundamental laws include at least whatever our best current physics gives us strong reason to think governs our universe.

For the 1st level layer, we now consider worlds that are 1) ours-like, and 2) permit simulations of ours-like world and sim observers (i.e., do not contain any law-profile sim blockers, substrate independence is valid, etc.).

For the 2nd layer, we first fix $N_{\text{sim}}(L, h)$ to be the number of ours-like simulated observers in $s = (L, h)$, and $N_{\text{non}}(L, h)$ the number of corresponding non-simulated observers. Then consider the different possible realizations h_s of such ours-like worlds as special cases. (ordered by $N_{\text{sim}}(L, h)$):

h_0 (**none**): $N_{\text{sim}}(L, h_0) = 0$

h_1 (**few**): $0 < N_{\text{sim}}(L, h_1) \ll N_{\text{non}}(L, h_1)$

h_2 (**many**): $N_{\text{sim}}(L, h_2)$ is large and comparable to $N_{\text{non}}(L, h_2)$

h_3 (**dominant**): $N_{\text{sim}}(L, h_3) \gg N_{\text{non}}(L, h_3)$ (simulated observers dominate under the chosen weighting)

Under a standard sampling relative to the chosen weighting, one obtains for each such h :

$$P(\text{Sim} \mid L, h) = \frac{N_{\text{sim}}(L, h)}{N_{\text{sim}}(L, h) + N_{\text{non}}(L, h)}.$$

Thus, as the ratio of simulated to non-simulated observers increases, so does the conditional probability that the observer is simulated.

The h_3 case corresponds to the familiar dominance scenario discussed by Bostrom and Chalmers: conditional on such a realization h , a randomly sampled ours-like observer is likely to be simulated.

What matters is that this probability is conditional on both the law-profile L and the realization h . The observer-counting step therefore does not by itself determine a reasonable present day credence that we are simulated: it yields a high $P(\text{Sim} \mid L, h)$

only for those (L, h) in which (i) L permits simulations of ours-like worlds and (ii) the realization is dominance-type (h_3). But we lack any justified basis for how likely it is that ours-like worlds realize h_3 rather than h_0 , h_1 , or h_2 . Without assumptions about the probability of realizations h , the prevalence question remains underdetermined.

3.2 Revelation worlds and certainty from within

The previous case was prevalence-based: conditional on (L, h) , a high $P(\text{Sim} \mid (L, h))$ came from observer-counting and a weighting assumption. Here I isolate a different route: evidence-based credence, where high confidence in simulation is driven by internally accessible evidence.

This time we consider a set of possible worlds s such that $\text{Host}(s, w)$, i.e., the set of possible host worlds of our world w . For the 1st level, we consider law-profiles L_R such that 1) L_R permits simulations of our actual world w , and 2) L_R permits interventions that generate publicly checkable evidence E whose best explanation, among the live internal competitors available to agents inside our world w , is that our environment is simulated (e.g. a public, reproducible, methodologically stable interaction that is best explained by a host-level intervention). The details of what would E constitute of are left schematic here.

For the 2nd level layer, we consider realizations h_R for such L_R where simulating of our world w is actually implemented, and where successful revelation event occurs. For such $s = (L_R, h_R)$, we then get:

$$S_R := \{ s \in S \mid \text{Host}(s, w) \wedge \text{a successful revelation occurs} \},$$

In this scenario, the simulation hypothesis is not supported primarily by anthropic prevalence, but by explanation. Given E , the body of internally available evidence produced by the revelation event, the rate of simulation credence can approach 1 (near certainty) from within w :

$$P(\text{Sim} \mid (L_w, h_w)) \approx 1,$$

in the epistemic sense that Sim becomes the best explanation among live competitors available to agents inside w .

3.3 Case comparison in 2-level framework

In the *prevalence* case, we ask which realizations h of an ours-like L lead to many simulations and use the two-level framework to isolate set of possible worlds that could be the future versions of our world w . The considered host worlds in this case are ours-like - they share some fundamental laws defined by core law-profile L_0 .

In the *revelation* case, we use our framework to isolate a set of possible worlds that could be the host worlds of our world w . In this second case, the host world’s laws need not resemble ours.

4 Consequences of high simulation credence

Suppose now that the previous cases were in fact true and that agents in w have strong reasons to believe that w is simulated. This does not necessarily change the reality of w for its inhabitants, but it changes what their data can support. Even if a regularity is exceptionless in w , its status as a law is no longer safe: it may hold only because the host implements w that way, and could be revised by host-level intervention without any prior misdescription of w .

Methodologically, this makes host-intervention a live explanatory competitor. Deviations from established regularities need not always indicate hidden variables or observational error, they may instead reflect interference at the host level. In that way anomalies expand the space of live explanations beyond internal competitors, and regularities confirm candidate laws only conditional on the absence of such interference.

In this sense, the contrast between the two cases also matters.

In the *prevalence* case (Section 3.1), the intended hosts are ours-like worlds containing L_0 , so high simulation credence need not eliminate confidence that some deep constraints in w track constraints of the host as well.

In the *revelation* case (Section 3.2), where host laws can be arbitrary relative to L_0 and E makes intervention explanatorily live, the reassurance of law similarity is not valid and even our best-supported “fundamental” generalizations about w carry an added layer of contingency.

5 Conclusion

The two-level representation (L, h) separates nomic feasibility from contingent realization and thereby disentangles two different types of questions relevant to simulation hypotheses: prevalence and discoverability. On the prevalence route, we explored an observer-counting case yielding a high $P(\text{Sim} \mid L, h)$ conditional on a dominance-type realization, leaving our present credence underdetermined.

On the discoverability route, we considered a revelation case, where host-world intervention can supply publicly checkable evidence E such that simulation becomes the uniquely best explanation from within w . For those cases (and possibly similar cases too), the hypothesis has a distinctive methodological upshot: it makes the apparent laws of w level-relative and potentially host-contingent.